

# Practice Questions for 2024-11-07-NetSimplex

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## 1 Questions

1. In the Network Simplex Method, what is the significance of a spanning tree  $T = (N, \tilde{A})$  in relation to the initial primal flow?
  - a. The spanning tree defines the arcs with non-zero flow in the initial primal flow, while all other arcs have zero flow.
  - b. The spanning tree is irrelevant to the initial primal flow; any feasible flow can be used as a starting point.
  - c. The spanning tree determines the number of iterations required to reach the optimal solution.
  - d. The spanning tree defines the set of nodes in the network, while the arcs are used to determine the initial flow.
  - e. The spanning tree is only used in the dual network simplex method, not the primal method.
2. In the context of the Dual Network Simplex Method, why is it crucial that the entering arc connects the two subtrees in the opposite direction of the leaving arc?
  - a. To ensure that the primal solution remains feasible after the pivot.
  - b. To guarantee that the dual solution remains feasible after the pivot.
  - c. To minimize the number of iterations required to reach the optimal solution.
  - d. To simplify the calculations involved in updating the dual variables.
  - e. This condition is not strictly necessary; the algorithm can function without it.
3. The Primal Network Simplex method starts with a spanning tree. If we add one arc to this spanning tree, what is the inevitable consequence, and how does this relate to the algorithm's operation?
  - a. The network becomes disconnected.
  - b. The spanning tree remains unchanged.
  - c. A cycle is created.
  - d. The optimal solution is immediately found.
  - e. The dual variables become infeasible.
4. In the Dual Network Simplex Method, what condition on the dual slack variables indicates that the current solution is not primal feasible, and what action is taken to address this?

- a. If all dual slack variables are positive, the solution is primal feasible.
  - b. If any dual slack variables are negative, the solution is not primal feasible, and a leaving arc is selected.
  - c. If any dual slack variables are zero, the solution is optimal.
  - d. The dual slack variables are irrelevant to primal feasibility.
  - e. If all dual slack variables are zero, the solution is not primal feasible.
5. In the Primal Network Simplex method, if we add one arc to the spanning tree  $T$ , what is the inevitable consequence, and how does this relate to the algorithm's operation?
- a. The spanning tree remains unchanged.
  - b. A cycle is created in the network.
  - c. The tree becomes disconnected.
  - d. The objective function value remains unchanged.
  - e. The dual variables are updated.
6. In the Primal Network Simplex method, if the entering arc connects two disjoint trees, what is the necessary condition for the leaving arc selection to maintain dual feasibility, and why?
- a. The leaving arc must be oriented in the same direction as the entering arc.
  - b. The leaving arc must be oriented in the opposite direction of the entering arc and have the smallest flow among all such arcs.
  - c. The leaving arc can be any arc in the cycle formed by adding the entering arc.
  - d. The leaving arc must have the largest flow in the cycle formed by adding the entering arc.
  - e. The leaving arc selection does not affect dual feasibility in the primal simplex method.
7. In the Primal Network Simplex method, if the entering arc connects two disjoint trees, what is the necessary condition for the leaving arc selection to maintain dual feasibility, and why?
- a. The leaving arc must be oriented in the same direction as the entering arc.
  - b. The leaving arc must be oriented in the opposite direction of the entering arc and have the smallest flow among all such arcs.
  - c. The leaving arc can be any arc in the cycle formed by the entering arc.
  - d. The leaving arc must have the largest flow in the cycle formed by the entering arc.
  - e. The leaving arc selection does not affect dual feasibility.

## 2 Answers

1. In the Network Simplex Method, what is the significance of a spanning tree  $T = (N, \tilde{\mathcal{A}})$  in relation to the initial primal flow?
  - a. A spanning tree provides a basis for the network flow problem. The arcs in the spanning tree ( $\tilde{\mathcal{A}}$ ) correspond to basic variables, and their flows are determined by solving the system of equations  $B\tilde{x}_B = \tilde{b}$ . Arcs not in the spanning tree have zero flow.
  - b. Incorrect. The spanning tree is crucial for defining the initial basic feasible solution (though it might not be primal feasible initially).
  - c. Incorrect. The spanning tree is a structural element; the number of iterations depends on the algorithm and the problem's characteristics.
  - d. Incorrect. The spanning tree defines both the nodes and a subset of arcs forming a connected acyclic subgraph.
  - e. Incorrect. The spanning tree is fundamental to both the primal and dual network simplex methods.
2. In the context of the Dual Network Simplex Method, why is it crucial that the entering arc connects the two subtrees in the opposite direction of the leaving arc?
  - a. Incorrect. The primal solution might become infeasible, but the primary concern in the dual simplex method is maintaining dual feasibility.
  - b. If the entering arc were in the same direction as the leaving arc, the dual slack of the leaving arc would become negative after the pivot, violating dual feasibility. The opposite direction ensures dual feasibility is maintained.
  - c. Incorrect. While it contributes to efficiency, the primary reason is maintaining dual feasibility.
  - d. Incorrect. The condition simplifies the update, but the main reason is to preserve dual feasibility.
  - e. Incorrect. This condition is essential for the correctness and convergence of the Dual Network Simplex Method.
3. The Primal Network Simplex method starts with a spanning tree. If we add one arc to this spanning tree, what is the inevitable consequence, and how does this relate to the algorithm's operation?
  - a. Incorrect. Removing an arc disconnects the tree; adding an arc creates a cycle.
  - b. Incorrect. Adding an arc changes the spanning tree structure.
  - c. Adding an arc to a spanning tree creates a cycle. This cycle is crucial because it determines how the flow is adjusted during a pivot operation in the primal simplex method. The algorithm then identifies a leaving arc from this cycle to maintain the spanning tree structure.
  - d. Incorrect. The algorithm iteratively improves the solution; adding an arc is just one step.
  - e. Incorrect. The primal simplex method focuses on primal feasibility; dual feasibility is addressed in the dual simplex method.
4. In the Dual Network Simplex Method, what condition on the dual slack variables indicates that the current solution is not primal feasible, and what action is taken to address this?
  - a. Incorrect. Positive dual slacks are a condition for dual feasibility, but primal feasibility requires non-negative flows.

- b. The Dual Network Simplex method starts with a dual feasible solution. A negative dual slack variable indicates a negative flow in the primal problem, violating primal feasibility. The algorithm then selects a leaving arc (the arc with the negative flow) and an entering arc to restore primal feasibility while maintaining dual feasibility.
  - c. **Incorrect.** Zero dual slacks are a condition for optimality, but primal feasibility is a separate condition.
  - d. **Incorrect.** Dual slack variables are directly related to primal flows through complementary slackness.
  - e. **Incorrect.** All zero dual slacks indicate optimality, not necessarily primal infeasibility.
5. In the Primal Network Simplex method, if we add one arc to the spanning tree  $T$ , what is the inevitable consequence, and how does this relate to the algorithm's operation?
- a. **Incorrect.** Adding an arc to a spanning tree always creates a cycle.
  - b. Adding an arc to a spanning tree creates a cycle. This is a fundamental property of trees. The algorithm uses this cycle to identify a leaving arc and update the flow, improving the solution.
  - c. **Incorrect.** Removing an arc disconnects the tree; adding an arc creates a cycle.
  - d. **Incorrect.** Adding an arc and updating the flow generally changes the objective function value.
  - e. **Incorrect.** While dual variables are updated, the creation of a cycle is the immediate consequence of adding an arc to a spanning tree.
6. In the Primal Network Simplex method, if the entering arc connects two disjoint trees, what is the necessary condition for the leaving arc selection to maintain dual feasibility, and why?
- a. **Incorrect.** Choosing a leaving arc in the same direction as the entering arc would violate dual feasibility.
  - b. The correct answer is B. If the leaving arc is oriented in the same direction as the entering arc, the dual slack of the leaving arc will become negative after the pivot, violating dual feasibility. Choosing the arc with the smallest flow ensures that the flow adjustment along the cycle does not lead to negative flows on other arcs.
  - c. The correct answer is C. In the dual simplex method, the starting solution is dual feasible but may not be primal feasible. Negative flow variables indicate primal infeasibility. The algorithm then selects an entering arc to increase the flow and restore primal feasibility.
  - d. **Incorrect.** This would not guarantee dual feasibility.
  - e. **Incorrect.** The leaving arc selection is crucial for maintaining dual feasibility in the primal simplex method.
7. In the Primal Network Simplex method, if the entering arc connects two disjoint trees, what is the necessary condition for the leaving arc selection to maintain dual feasibility, and why?
- a. **Incorrect.** If the leaving arc is oriented in the same direction as the entering arc, the dual slack of the leaving arc will become negative after the pivot, violating dual feasibility.
  - b. The correct answer is B. To maintain dual feasibility, the leaving arc must be oriented in the opposite direction of the entering arc within the cycle created by adding the entering arc. If the leaving arc were oriented in the same direction, its dual slack would become negative after the pivot, violating dual feasibility. Selecting the arc with the smallest flow among those oriented in the opposite direction ensures that the flow remains non-negative after the update.

- c. Incorrect. The leaving arc cannot be any arc in the cycle; it must be oriented in the opposite direction of the entering arc to maintain dual feasibility.
- d. Incorrect. Selecting the arc with the largest flow does not guarantee dual feasibility; the orientation of the arc is crucial.
- e. Incorrect. The leaving arc selection directly impacts dual feasibility; an incorrect choice can lead to a dual infeasible solution.